RE.AI SOLUTIONS

AI-Powered Virtual Staging

Presented by Brandon Matias

The Problem in Staging

Key Challenges in Real Estate Marketing

Empty homes tend to sell **slowly and for less**, creating a significant barrier for sellers. Traditional staging is expensive and current digital tools often lack reliability and affordability.

Introducing RE.AI Solution

Transforming Empty Spaces into Homes

RE.AI Solution revolutionizes real estate staging by providing a web-based platform that converts empty room photos into beautifully furnished images instantly, ensuring **100% architectural integrity** and market readiness.

Technical Architecture

Overview of the system components and AI engines

Frontend System

The frontend is built on Replit, utilizing GCS-backed Object Storage for efficient file handling and user interactions.

Database Management

PostgreSQL (Neon) serves as the database, managing job tracking and token usage to ensure smooth operations and performance.

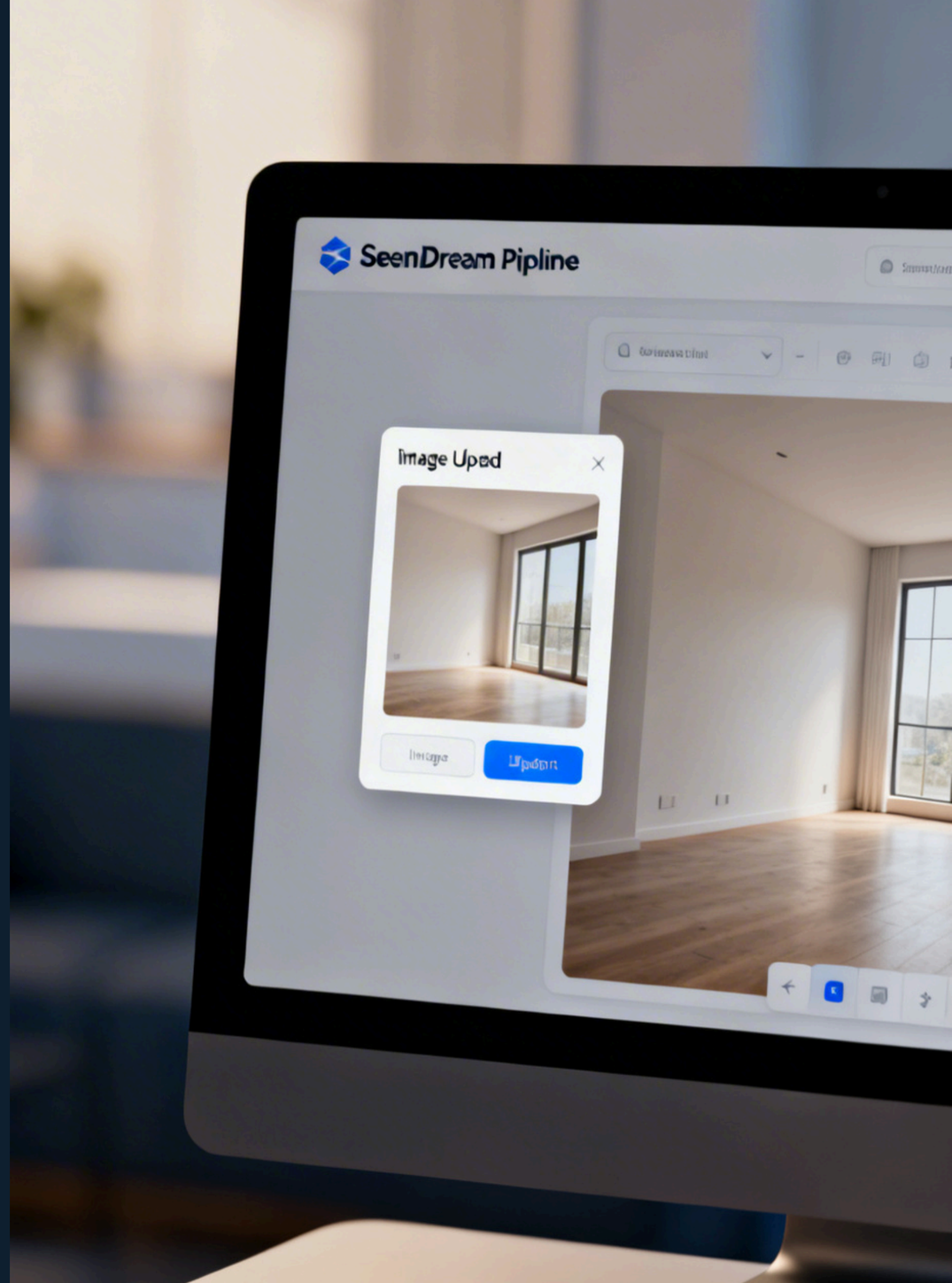
AI Integration

The architecture incorporates primary and fallback AI engines, optimizing image processing capabilities while maintaining reliability and performance.

Data Plan

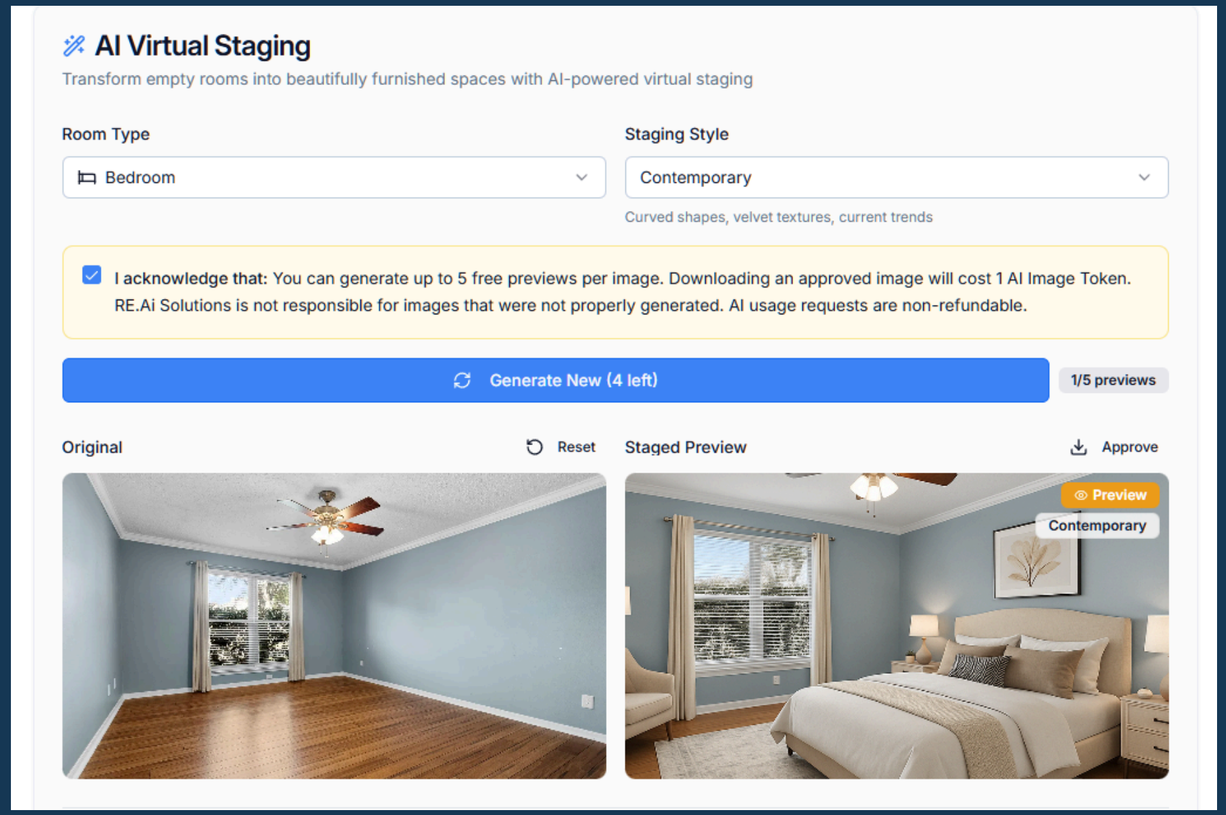
Under the hood deep dive

- **Source:** Custom-curated library of 30+ style-specific prompts for architectural consistency.
- **Input Data:** High-resolution room photos (JPG, PNG, WebP) converted to signed URLs via Replit Object Storage.
- **Size:** Tested on 6 primary indoor room types (Living, Bedroom, Kitchen, Bathroom, Dining, Office) and specialized outdoor/garage categories.
- **Style Variants:** Each room type linked to 5 design styles: Modern, Contemporary, Rustic, Minimalist, Traditional.
- **Labels/Metadata:** Staging requests tagged with a unique Job ID in a PostgreSQL (Neon) database to track room types and styles.



System Design

Key Concepts for Quality and Integrity in AI



Style Matrix

The **style matrix** offers over 30 combinations, facilitating diverse room types and styles for accurate staging representations across various settings.

Additive Patterns

Our **additive patterns** effectively ignore walls, windows, and floors, ensuring furniture is placed seamlessly without altering the room's original structure.

Integrity Rules

Established **structural integrity rules** prevent architectural hallucinations, ensuring realistic outcomes that maintain the original design and layout of the space.

Success Metrics

Prove the solution works and is reliable for professional real estate use.

- **Primary Metric:** Structural Integrity (Accuracy): Aim for 0% errors in permanent architectural features like walls, windows, and floors.
- **Secondary Metric:** Latency (Speed): Deliver high-resolution (1024x1024) images in under 15 seconds per request.
- **Qualitative Metric:** Visual Realism: Ensure furniture appears grounded with natural shadows using "Additive Staging Patterns".
- **System Reliability:** Maintain 100% uptime with an automated fallback to OpenAI's image API if needed.
- **User Engagement:** Track success with the Token System by measuring how many "Preview" generations lead to final user "Approval".

Project Roadmap

Evaluating RE.AI Solution's History

Phase 1: Foundation & Infrastructure (Dec 2025) establishes the base with AI image enhancement and launches the "Virtual Staging Showcase" on December 3–4. It also implements pay-per-download models and security features by December 5.

Phase 2: Structural Accuracy Sprint (Jan 12, 2026) focuses on precision with ControlNet for better 3D perspective, "Structural Preservation" to maintain room features, and "Forbidden Furniture" lists for style refinement.

Phase 3: Supporting Ecosystem (Jan 19, 2026) introduces Agent Portfolios for users to manage and display staged results.

Phase 4: Optimization & Final Pipeline (Jan 20, 2026) includes migrating to SeeDream 4.5 for architectural consistency and revamping the backend for efficiency, finalizing the Additive Staging Pattern, and specialized logic for garages and outdoor spaces.

Challenges & Mitigations

Addressing potential issues and proposed solutions effectively

Model Downtime - Medium

Automated fallbacks to OpenAI API ensure uninterrupted service during model downtimes, maintaining a seamless user experience without delays.

Floating Artifacts - Low

Implementing refined structural integrity rules minimizes floating furniture artifacts, ensuring realistic staging while preserving the appearance of natural shadows.

Architectural Hallucinations - Low

Strict adherence to structural integrity rules prevents architectural hallucinations, avoiding legal risks and ensuring that generated images retain authenticity and accuracy.

Resources Needed

Breakdown of resources, frameworks, and costs.

Compute Resources

Leveraging Replicate API and Replit for hosting will ensure efficient processing and seamless user experiences as image demand grows.

Framework Implementation

Utilizing Hugging Face with Node.js/Express and Neon/PostgreSQL will enhance performance and scalability for comprehensive AI functionalities.

Cost Estimation

Projecting a $5-$10 cost per 100 images allows for sustainable growth while providing valuable services and robust features to users.

Thank You!

See it in action at:
www.reaisolution.com